2023. M81A 2023L562A1EL

Coimisiún na Scrúduithe Stáit
State Examinations Commission

Leaving Certificate Examination, 2023

Design & Communication Graphics
Higher Level
Section A (60 marks)

Centre No.

Thursday, 22 June
Morning, 9:30 - 12:30

This examination is divided into three sections:

SECTION A	(Core - Short Questions)
SECTION B	(Core - Long Questions)
SECTION C	(Applied Graphics - Long Questions)

SECTION A
- Four questions are presented.
- Answer **any three** questions on the A3 sheet overleaf.
- All questions in Section A carry **20 marks** each.

SECTION B
- Three questions are presented.
- Answer **any two** questions on drawing paper.
- All questions in Section B carry **60 marks** each.

SECTION C
- Five questions are presented.
- Answer **one** question (i.e. the option you have studied) on drawing paper.
- All questions in Section C carry **60 marks** each.

General Instructions:
- *Construction lines must be shown on all solutions.*
- *Write the question number distinctly on the answer paper in Sections B and C.*
- *Work on one side of the drawing paper only.*
- *All dimensions are given in metres or millimetres.*
- *Write your Examination number in the box below and on all other sheets used.*

Examination Number

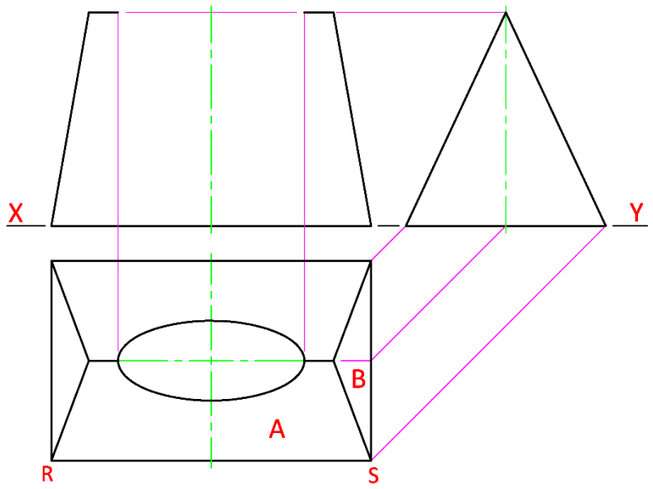
SECTION A - Core - Answer **any three** of the questions on this A3 sheet.

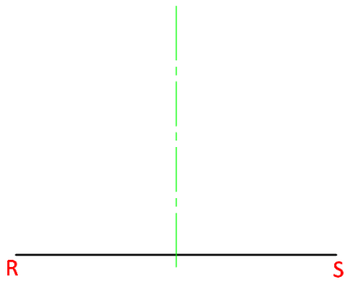
A-1. The image on the right shows a tissue box. The opening at the top of the box is elliptical in plan.

The drawing below shows the plan, incomplete elevation and end view of a similar tissue box. The position of a surface development of surface **A** is also shown.

(a) Complete the elevation.

(b) Complete a one-piece surface development of surfaces **A** and **B**.

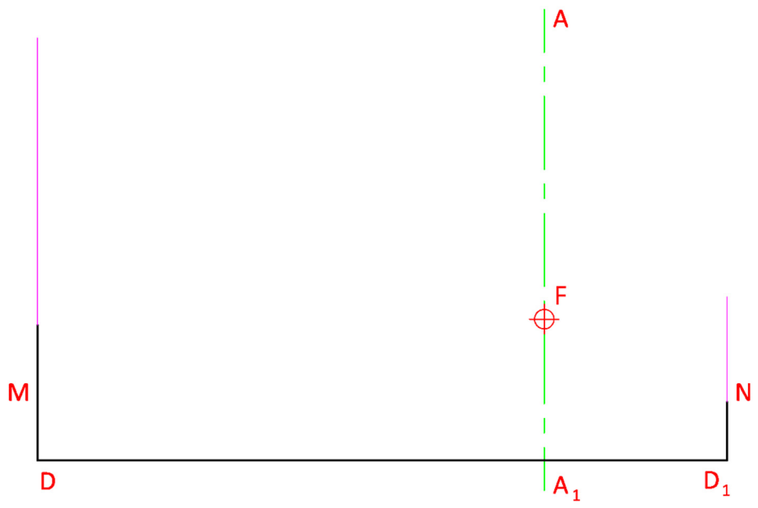




A-2. The image below shows the ski jump from the 2022 Winter Olympics in Beijing China. The ski jump slope is in the shape of a hyperbola. The drawing shows the axis **AA$_1$**, directrix **DD$_1$**, focus **F**, and incomplete edges **M** and **N** of a similar ski jump. The hyperbola has an eccentricity of 3:2.

(a) Locate the line of eccentricity and vertex of the hyperbola.

(b) Draw the portion of the hyperbola between edges **M** and **N** and complete the ski jump.

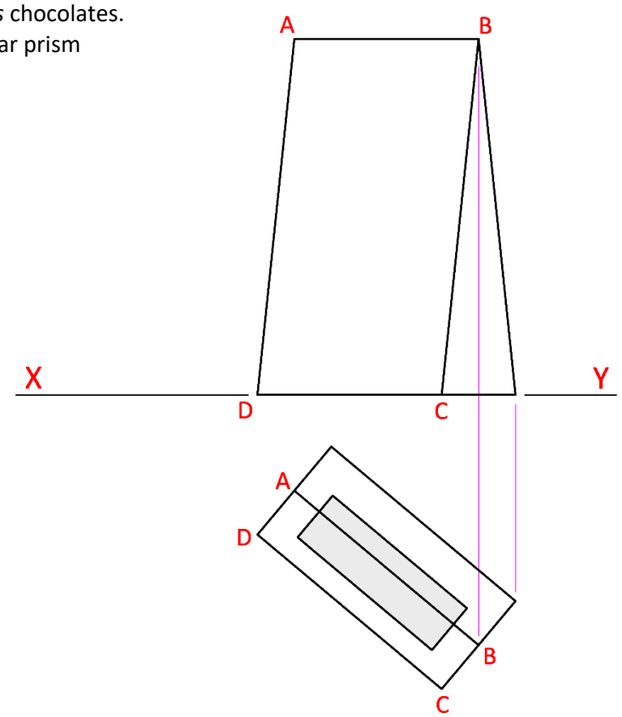


A-3. The image below shows the packaging for *Skelligs* chocolates. The shape of the packaging is based on a triangular prism with a rectangular label at the top.

The drawing on the right shows the plan and incomplete elevation of a similar package.

(a) Determine the horizontal and vertical traces of the oblique plane that contains the surface **ABCD**.

(b) Complete the elevation of the label (shaded area in plan) including all hidden detail.

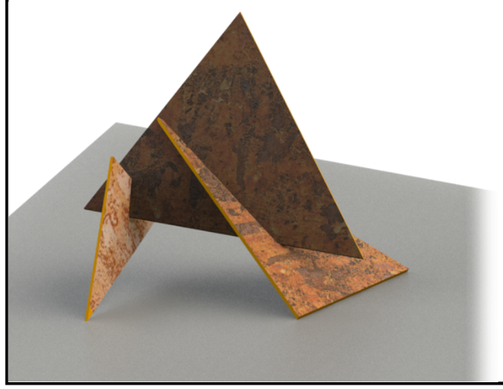


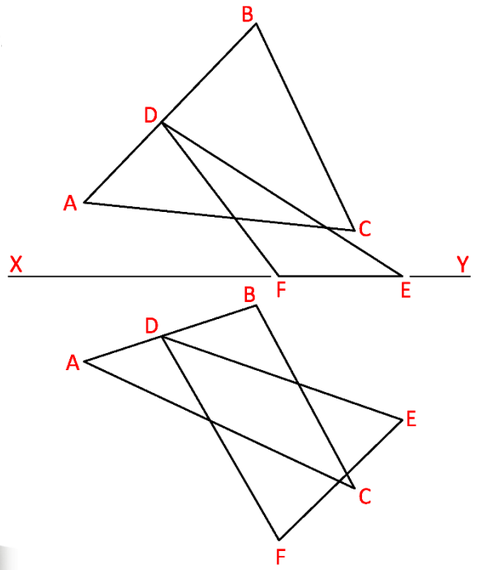
A-4. The image below shows a sculpture based on intersecting triangular planes

The drawing on the right shows the incomplete plan and elevation of two of these intersecting planes.

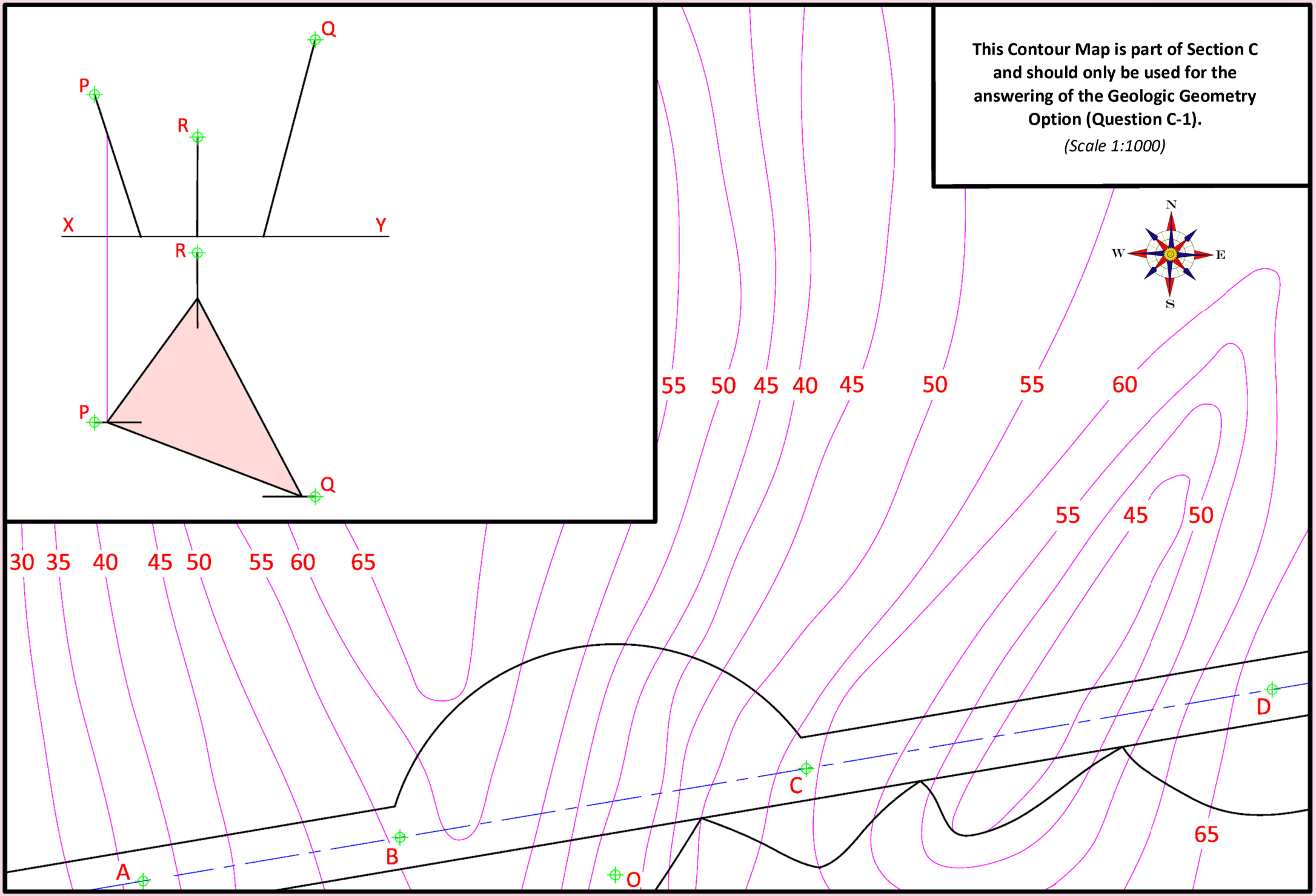
(a) Draw the plan and elevation of the line of intersection between the two planes.

(b) Determine the true shape of the plane **DEF**.





This examination paper must be returned at the end of the examination – You must include your Examination Number on the front cover.


Q
P
R
X
Y
R
P
Q
This Contour Map is part of Section C and should only be used for the answering of the Geologic Geometry Option (Question C-1).
(Scale 1:1000)
N
W
E
S
55 50 45 40 45 50 55 60
55 45 50
30 35 40 45 50 55 60 65
D
C
65
A
B
O